

Breaking the Manual Alignment Barrier: First Demonstration of an Automatically Aligned Kilometer-Scale 100-Gb/s Photonic THz Wireless Link in 300-GHz Band

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Abstract We demonstrate the first automatically aligned km-scale 100-Gb/s photonic THz wireless link under realistic rooftop-to-rooftop deployment conditions. Sensor-assisted coarse acquisition and power-feedback fine alignment achieve 25-GBaud 16-QAM transmission over a 1.1-km line-of-sight path at 300 GHz with a BER of 1.5×10^{-2} . ©2026 The Author(s)

Introduction

As mobile networks evolve toward 6G and beyond, fronthaul and backhaul links increasingly demand fiber-like throughput [1, 2]. The terahertz (THz) spectrum, offering bandwidths of tens of gigahertz, has emerged as a promising wireless alternative where fiber deployment is costly or impractical [3-5]. Recent advances in photonics-based THz signal generation have enabled THz wireless links ranging from a few meters to the kilometer scale [6-11].

However, extending THz links to km-scale outdoor distances exposes a critical challenge that remains largely unaddressed: beam alignment. At 300 GHz, the narrow beamwidth required to overcome severe free-space path loss and atmospheric absorption makes the link highly sensitive to angular misalignment. Figure 1 summarizes representative THz wireless link demonstrations. Prior long-range high-data-rate THz demonstrations, including links 500 m and beyond, have relied on manually aligned terminals, limiting reproducibility and rendering such systems impractical for real-world deployment [12-14]. In parallel, beam-control demonstrations for THz communications have mostly remained at short ranges, from sub-meter and meter-scale beam-steering links to, more recently, sensing-assisted mobile operation over tens of meters [15-20]. Prior to this work, automatic end-to-end beam alignment had not been demonstrated for THz wireless links in an outdoor km-scale regime.

Our previous work demonstrated automatic fine beam alignment using power-feedback raster scanning in a 200-m photonic THz link at 300 GHz [21], with initial pointing performed manually. At km-scale, manual initial pointing becomes impractical: the opposite terminal is no longer readily visible, and the narrower antenna beamwidth demands pointing precision beyond what manual pointing can reliably achieve. Automating the initial acquisition stage — where neither terminal knows the other's direction a priori — is therefore essential for the entire alignment chain to operate without manual intervention.

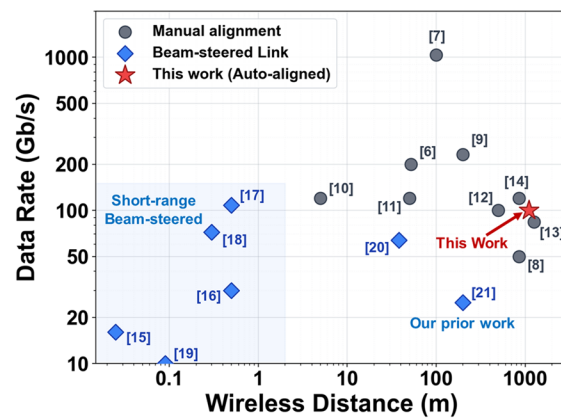


Fig. 1: State-of-the-art THz wireless link demonstrations.

Here, we realize this by integrating sensor-assisted initial beam acquisition with power-feedback fine alignment, thereby forming an automatic alignment chain from initial pointing to final convergence. Each terminal's position and heading are obtained from a dual-antenna global navigation satellite system (GNSS) receiver, while an inertial measurement unit (IMU) provides tilt information, enabling coarse pointing with sufficient accuracy for subsequent fine alignment. With this approach, we demonstrate, to the best of our knowledge, the first 100-Gb/s photonic THz wireless link with automatic end-to-end (coarse + fine) beam alignment at km-scale distance. Specifically, a single-carrier 16-QAM signal is successfully transmitted over a 1.1-km rooftop-to-rooftop line-of-sight (LOS) path at 300 GHz, achieving a BER of 1.5×10^{-2} , below the 2×10^{-2} threshold of 20% soft-decision forward error correction (SD-FEC).

Sensor-Assisted Coarse Acquisition and Power-Based Fine Alignment

As illustrated in Fig. 2, the goal of coarse acquisition is to point both the Tx and Rx antennas toward each other with sufficient accuracy to enter the fine-alignment capture range. This is achieved by combining dual-antenna GNSS position and heading information with IMU tilt measurements. Each terminal obtains its geodetic

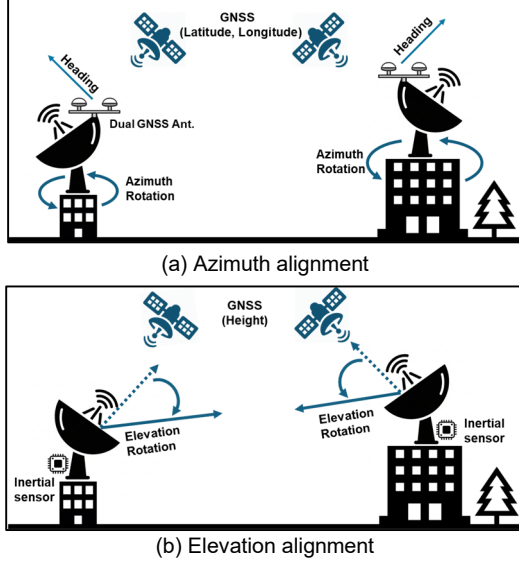


Fig. 2: Conceptual illustration of sensor-assisted coarse acquisition for initial azimuth and elevation alignment.

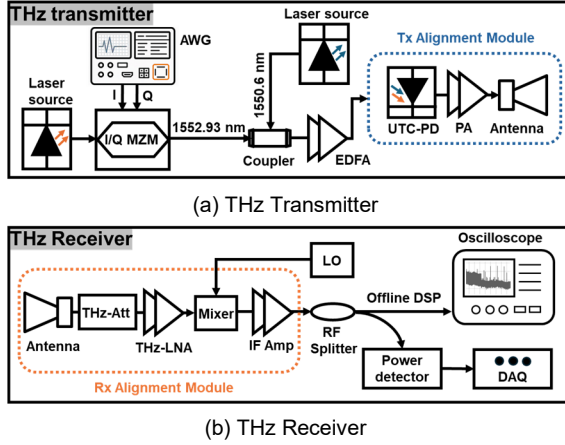


Fig. 3: System configuration of the photonic THz link.

coordinates — latitude ϕ , longitude λ , and altitude h — from the GNSS receiver. Given the Tx position (ϕ_T, λ_T, h_T) and Rx position, (ϕ_R, λ_R, h_R) the relative azimuth, θ_{az} from Tx toward Rx is

$$\theta_{az} = \arctan2(y, x) \text{ where } y = \sin \Delta\lambda \cos \phi_R, \quad (1)$$

$$x = \cos \phi_T \sin \phi_R - \sin \phi_T \cos \phi_R \cos \Delta\lambda$$

where $\Delta\lambda = \lambda_R - \lambda_T$. The reverse azimuth from Rx toward Tx is obtained by swapping the subscripts. The horizontal distance d , computed via the Haversine formula, yields the elevation angle:

$$\theta_{el} = \arctan((h_R - h_T)/d) \quad (2)$$

The dual-antenna GNSS provides the antenna boresight heading θ_{head} , and the required azimuth steering angle is then simply $\Delta\theta_{az} = \theta_{az} - \theta_{head}$. For elevation, the IMU measures the current tilt angle θ_{tilt} of the antenna assembly, yielding the required elevation steering angle $\Delta\theta_{el} = \theta_{el} - \theta_{tilt}$. After these corrections are applied at both terminals, the antennas are coarsely pointed toward



Fig. 4: Photograph of the deployed link (top) and map of the 1.1-km path between ETRI and KAIST (bottom).

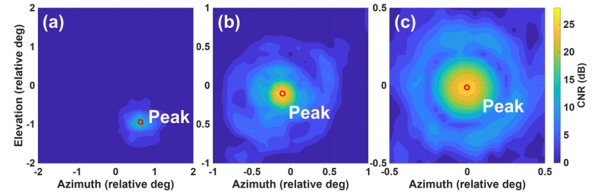


Fig. 5: Measured CNR maps of the Tx raster scan at each alignment stage: (a) after coarse pointing, (b) after the first fine-alignment round, (c) after the second fine-alignment round.

each other. If the residual pointing error falls within the fine-alignment capture range, which is defined by the raster scan field of view, the subsequent fine-alignment stage can be initiated.

After coarse acquisition, the antennas are pointed within the capture range but not yet at peak alignment. Fine alignment refines the pointing by a two-dimensional raster scan centered on the coarse-pointed direction, maximizing the received IF power at the Rx. Throughout the scan, the peak power and its angular position are held (max-hold), and the antenna is then steered to that stored position. The Tx and Rx terminals perform fine alignment alternately: one terminal scans while the other remains stationary, then the roles are reversed. This avoids the ambiguity that would arise if both antennas moved simultaneously. The scan is performed in two stages—first a wide raster with large step size to locate the approximate beam center, followed by a narrow raster with smaller step size around that position—achieving sub-beamwidth pointing precision without scanning the entire window at fine resolution.

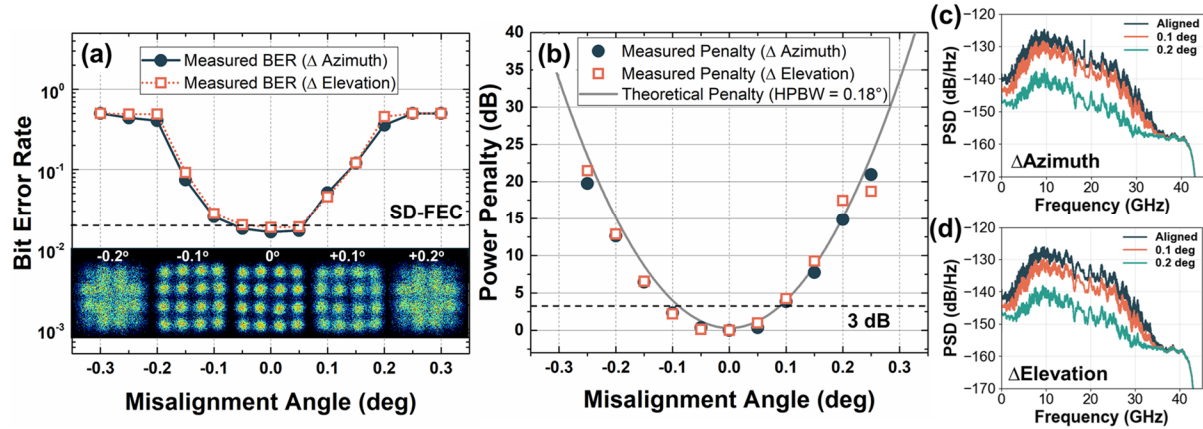


Fig. 6: (a) Measured BER vs. misalignment angle at the Rx, with constellation insets; (b) power penalty vs. misalignment angle with theoretical Gaussian-beam (HPBW = 0.18°) fit; received signal spectra at 0.1° and 0.2° offsets in (c) azimuth and (d) elevation.

Experimental setup and results

Figure 3 shows the experimental setup. At the Tx, the 300-GHz-band carrier (290 GHz) is generated by heterodyning two optical tones in a UTC-PD, one of which carries a 25-GBaud 16-QAM waveform from an IQ Mach-Zehnder modulator. The resulting THz signal is amplified to +2 dBm and radiated through a Cassegrain antenna with ~55 dBi directivity. At the receiver, a matched antenna collects the incoming signal, which is amplified by a THz-LNA and down-converted by a subharmonic mixer to an 18.5-GHz IF, captured by a real-time oscilloscope at 256 GSa/s for offline DSP. In parallel, a power detector continuously monitors the received IF power during beam alignment. Both antennas are mounted on two-axis motorized stages with dual-antenna GNSS receivers and IMUs for coarse acquisition as described above. The link spans a 1.1-km rooftop-to-rooftop LOS path between KAIST and ETRI in South Korea as in Fig. 4, with estimated free-space path loss of 142.5 dB at 290 GHz.

Figure 5 shows the measured CNR maps of the Tx raster scan at each alignment stage. As illustrated in Fig. 5(a), immediately after coarse pointing, a wide $\pm 2^\circ$ scan reveals a power peak offset by approximately 0.5° in azimuth and 1° in elevation from the scan center. The residuals exceed the nominal sensor accuracies ($\sim 0.12^\circ$ heading, $\sim 0.3^\circ$ tilt), mainly due to GNSS positioning uncertainty and mechanical misalignment between the sensor reference frames and the RF boresight. In azimuth, the residual is additionally affected by GNSS heading multipath from the nearby Cassegrain reflector. Nevertheless, the coarse-pointing error remains within the $\pm 2^\circ$ capture range, allowing the subsequent fine-alignment stage to locate the peak without manual intervention. In Fig. 5(b) and (c), two rounds of alternating Tx-Rx fine alignment with progressively narrower scan windows and step sizes bring the peak to the scan center. Sensor-assisted coarse

acquisition combined with power-feedback-based fine alignment thereby enables automatic alignment over the 1.1-km link.

To characterize the misalignment tolerance of the link, Fig. 6(a) plots the measured BER as a function of angular offset in azimuth and elevation at the Rx. The BER remains below the SD-FEC threshold only within $\pm 0.05^\circ$, and exceeds it at just 0.1° of offset, highlighting the stringent alignment accuracy required for reliable km-scale THz links. Figure 6(b) also plots the measured power penalty as a function of misalignment angle. Even a 0.2° offset incurs a power penalty approaching 15 dB, further confirming the sub- 0.1° alignment precision demanded at km-scale distances. The theoretical penalty curve, calculated from a Gaussian beam profile with an estimated half-power beamwidth (HPBW) of 0.18° , is overlaid and closely matches the measured data, confirming the effective beamwidth of the employed Cassegrain antenna. The 3-dB penalty boundary falls at approximately $\pm 0.09^\circ$, well within the precision achieved by the proposed automatic alignment system. Figures 6(c) and (d) show the received signal spectra at 0.1° and 0.2° offsets in azimuth and elevation, respectively. The spectra remain flat across the bandwidth with only a uniform power reduction at both 0.1° and 0.2° offsets, confirming that the misalignment penalty is dominated by antenna gain roll-off rather than frequency-selective distortion.

Conclusions

We have demonstrated the first automatically aligned km-scale photonic THz wireless link operating at 300 GHz. By combining sensor-assisted coarse acquisition with power-feedback fine alignment, automatic alignment was achieved over a 1.1-km LOS path, enabling 100-Gb/s 16-QAM transmission with a BER of 1.5×10^{-2} . The results confirm that automatic alignment at km-scale THz distances is feasible, removing a key barrier to practical outdoor deployment.

Acknowledgements

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